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CLINICAL AT

Serum Biomarkers Identify Patients Who Will Develop Inflammatory Bowel Diseases Up to 5 Years Before Diagnosis

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BACKGROUND & AIMS: Biomarkers are needed to identify patients at risk for development of inflammatory bowel diseases. We aimed to identify serum biomarkers of Crohn's disease and ulcerative colitis that can be detected and quantified before diagnosis. **METHODS:** We obtained serum samples from patients archived before a diagnosis of Crohn's disease (n = 200) or ulcerative colitis (n = 199), as well as from 200 healthy individuals (controls), collected from 1998 through 2013 as part of the US Defense Medical Surveillance System. We measured levels of antibodies against microbes (anti–*Saccharomyces cerevisiae* IgA or IgG, anti–*Escherichia coli* outer membrane porin C, anti-CBir1, anti-flagellin 2, anti-flagellin X, and perinuclear anti-neutrophil cytoplasmic antibodies) and 1129 proteins in each sample. We then used functional principal component analysis to derive the time-varying trajectory for each marker, which then was used in a multivariate model to predict disease status. Predictive performances at different prediagnosis timepoints were evaluated using area under the receiver operating characteristic curves (AUROCs). Biological pathways that were up-regulated in serum from patients with Crohn's disease were identified based on changes in protein abundance at different time periods preceding diagnosis. **RESULTS:** We identified a panel of 51 protein biomarkers that were predictive of Crohn's disease within 5 years with an AUROC of 0.76 and a diagnosis within 1 year with an AUROC of 0.87. Based on the proteins included in the panel, imminent development of CD was associated with changes in the complement cascade, lysosomes, innate immune response, and glycosaminoglycan metabolism. Serum antibodies and proteins identified patients who received a diagnosis of ulcerative colitis within 5 years with an AUROC of only 0.56 and within 1 year with an AUROC of 0.72. **CONCLUSIONS:** We identified a panel of serum antibodies and proteins that were predictive of patients who will receive a diagnosis of Crohn's disease within 5 years with high accuracy. By contrast we did not identify biomarkers associated with future diagnosis of ulcerative colitis.

Keywords: ASCA; IBD; pANCA; prognostic.

Inflammatory bowel diseases (IBD), including Crohn's disease (CD) and ulcerative colitis (UC), are chronic disorders, with increasing prevalence worldwide.[1] Because IBD is a chronic lifelong disease with onset at a young age, its prevalence is expected to sharply increase in the following decades.[2] IBD mainly affects young adults and may have devastating consequences on quality of life, health resource consumption, and work productivity.[2] Despite progress in our understanding of disease pathogenesis and major advances in therapeutic resources, there is still no cure for IBD, and drug-free remission is not possible in the majority of cases.[3] Furthermore, a considerable proportion of patients are refractory to or intolerant to therapies. More than ever, there is a need to develop novel strategies for earlier disease diagnosis and, ideally, disease prevention.

Preliminary evidence suggests that, like in other immune-mediated diseases, IBD is preceded by an asymptomatic preclinical period.[4,5] Gaining insight into this phase could allow disease prediction. The Proteomic Evaluation and Discovery in an IBD Cohort of Tri-service Subjects (PREDICTS) study was initiated with the goal of identifying biomarkers associated with IBD onset by using preclinical serum samples stored in the US Department of Defense Serum Repository (DoDSR). Here, we aimed to derive a predictive model for IBD that included hypothesis-driven

***Authors share co-first authorship; §Authors share co-senior authorship.**

Abbreviations used in this paper: **APCS, serum amyloid P component; ASCA, anti–*Saccharomyces cerevisiae* antibodies; CD, Crohn's disease; CRP, C-reactive protein; DMSS, Defense Medical Surveillance System; DoDSR, US Department of Defense Serum Repository; HC, healthy control; IBD, inflammatory bowel disease; ICD-9, *International Classification of Diseases, Ninth Revision*; IL, interleukin; KEGG, Kyoto Encyclopedia of Genes and Genomes; LASSO, least absolute shrinkage and selection operatory; pANCA, perinuclear anti–neutrophil cytoplasmic antibodies; PREDICTS, Proteomic Evaluation and Discovery in an IBD Cohort of Tri-service Subjects; UC, ulcerative colitis.**

Most current article

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WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

Tests are needed to identify patients at risk for development of inflammatory bowel diseases.

NEW FINDINGS

We identified a panel of serum antibodies and proteins that identifies patients who will receive a diagnosis of Crohn's disease within 5 y with high accuracy.

LIMITATIONS

These findings must be replicated in a separate, larger cohort of patients.

IMPACT

This panel of biomarkers might be used to identify patients at risk for Crohn's disease who should take measures to prevent disease development.

antibodies and protein biomarkers and to leverage sophisticated proteomic analysis to identify biological pathways that might plausibly drive and mediate the preclinical phase of IBD.

Methods

Study Design and Patients

This was a case-control study, nested within a cohort study, described elsewhere (Figure 1).[6] Briefly, cases with an incident diagnosis of CD and UC were identified between 1998 and 2013 within the Defense Medical Surveillance System (DMSS) and linked to the DoDSR.[7] Incident cases of IBD had 2 or more medical encounters with an *International Classification of Diseases, Ninth Revision* (ICD-9) code for CD (555.0, 555.1, and 555.9 [including all subgroup codes]) or UC (556 [including all subgroup codes]) and available serum from the time of IBD diagnosis (±1 year) and from the preceding sampling points. Date of IBD diagnosis was the first time an IBD ICD-9 code was registered.[6] Overall, 1000 CD cases, 1000 UC cases, and 500 healthy control (HC) cases were selected from the DMSS database following a prespecified algorithm published elsewhere[6] and described in the supplementary material. For this nested cohort study, 200 CD, 200 UC, and 200 HC cases were further selected as follows. Disease phenotype was ascertained as described previously.[5] From each individual, 3 or 4 serum samples were retrieved: 1 corresponding to the nearest sample available around the date of diagnosis (±1 year; sample A), and the other 2 or 3 (samples B, C, and D) corresponding to samples preceding the diagnosis of IBD and separated approximately by 2-year intervals (±6 months). Sample D was the earliest available specimen in the DoDSR. HC individuals were required to have no medical encounter with evidence of IBD, rheumatoid arthritis, celiac disease, or colon cancer (based on ICD-9 codes). Because of logistical constraints, samples were first obtained from patients with CD. Samples from HC individuals matched on age, sex, race, and timing of sample A (±1 year) to patients with UC were obtained at a later date. Further details and the sample distribution (Supplementary Figure 1) can be found in the supplementary material.

We further conducted a search for gastrointestinal-related complaints in the years preceding diagnosis, using the codes described in Supplementary Table 1 and represented in Supplementary Figure 2.

Sample Testing

Each sample was tested for a panel of circulating antibodies: perinuclear anti–neutrophil cytoplasmic antibodies (pANCA), anti–*Saccharomyces cerevisiae* antibodies (ASCA), IgA and IgG (directed against a cell wall polysaccharide of the yeast), anti–*Escherichia coli* outer membrane porin C (OmpC), and anti-flagellins antibodies (anti-CBir1, anti-flagellin 2, and anti-flagellin X). Testing for antibodies was conducted at Prometheus Laboratories (San Diego, CA).

Protein abundance was tested using SomaLogic (Boulder, CO), a multiplex platform profiling 1129 biomarkers (SOMAmers), representing a broad range of biological functions.[8]

Statistical Analysis

A detailed description of the analysis can be found in the supplementary material. After log transformation, antibody and SomaLogic data were batch corrected. Then, samples were randomly divided into training and testing data (Supplementary Table 2). Specifically, training data were used to select markers associated with CD and UC (separate models were estimated for each disease), and the testing subset was used to evaluate the predictive performance of the selected markers. Only samples obtained before day 0 of diagnosis for CD or UC were used.

Time-Varying Trajectory of the Markers. Because samples were irregularly distributed over time (because of serum availability in the DoDSR) and timepoints not well aligned, a 2-step strategy for data analysis was adopted. First, using markers' abundance available for each patient before diagnosis, functional principal component analysis was performed to estimate the trajectory through time of different markers for each patient.[9–11] Functional principal component analysis models the abundance of each marker as a function of time via a stochastic process, to capture the change along time of each marker. Principal component models were estimated based on samples in the training data, and for each sample in the training and testing data, the abundance of each marker was predicted over time (Supplementary Figure 3). Then, predicted markers' abundances at a particular time before diagnosis was used in a regression model to predict the disease of interest (ie, CD or UC) (see supplementary material).

Prediction Model. For each time T (ie, 1, 2, 3, 4, and 5 years) before diagnosis, disease status was modeled as a function of the estimated abundance of the 7 antibodies at time T based on the training data via the least absolute shrinkage and selection operator (LASSO) algorithm.[12] Given the high dimensionality of SomaLogic data, variable selection was performed to identify the most predictive markers based on the training data. Thereafter, disease status was modeled as a function of the abundance of selected SOMAmers at a given time Tbefore diagnosis via random forest.[13] Random forest was used for SomaLogic data, given the higher dimensionality of the data involved. Finally, the predictive performance of the selected markers was validated on the testing data, and ROC curves were derived for each year before diagnosis.

Risk of Developing Crohn's Disease. Besides assessing predictive performance, we were interested in

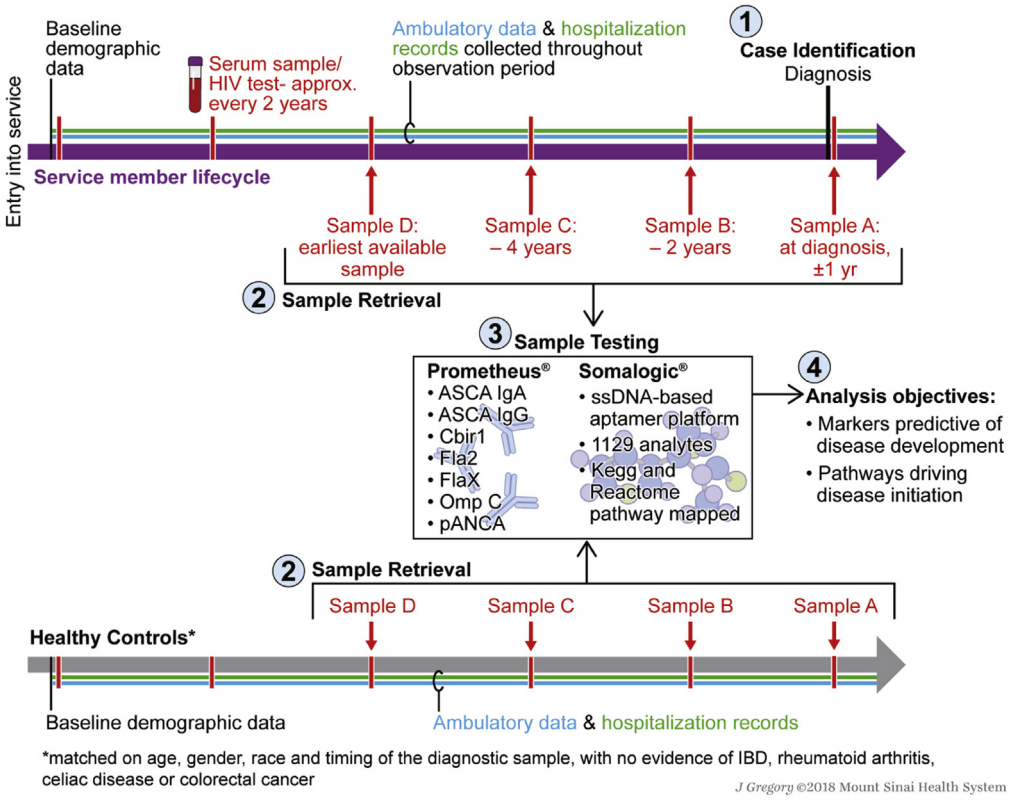


Figure 1. The PREDICTS study design. Incident cases of IBD were identified through a combination of administrative coding and linked to their serum samples in the DoDSR. From each patient with IBD and from HC individuals, 3 or 4 serum samples were retrieved. A panel of antibodies and proteins was measured with the goal of identifying markers predictive of disease development and pathways involved in the preclinical phase of disease. For the prediction model described in the article, only samples obtained before the day of diagnosis were used. ssDNA, single-strand DNA.

quantifying the risk of developing the disease of interest for each patient given the predicted markers' abundance at a particular timepoint before diagnosis. For each patient in the testing data, the risk of developing the disease was derived by comparing the abundance of markers with those estimated based on HC individuals and patients with CD in the training data. Bonferroni-adjusted *P* values for 1-sided Wilcoxon test were calculated. Basically, a higher risk of developing the disease was estimated if the marker's abundance for a particular sample was more comparable to that estimated for patients with CD than that derived for HC individuals (Supplementary Figure 4). For probability computation, the prevalence of IBD in the US Army (0.14%) was used.[14,15]

Pathway-Driven Prediction Model. Canonical pathways from the MSigDB database were considered.[16] Based on their predictive performance in a 5-fold cross-validation technique performed on the training data, SOMAmers were ordered from most to least predictive of CD, and pathways enriched at the top of the list were found by using the Kolmogorov-Smirnov statistics.[17,18] The significance of enriched pathways was assessed with permutation techniques.[17] Finally, the predictive performance of SOMAmers mapping to enriched pathways was evaluated on the testing data (Supplementary Material).[17]

Results

Population Demographics and Sample Characteristics

Two hundred patients with CD, 199 patients with UC, and 200 HC individuals were linked to their samples in the DoDSR (Table 1). A total of 1824 samples before diagnosis were available. The furthest sample in relation to the date of diagnosis (UC and CD) or timing of sample A (HC) was obtained at a median of −5.9 years (interquartile range [IQR], 0.47) for CD, −7.1 years (IQR, 2.4) for UC, and −6.4 years (IQR, 2.4) for HC. The presence of at least 1 gastrointestinal-related ICD-9 code was more frequent in individuals who developed CD or UC in the year preceding diagnosis but similar to control individuals when going back at least 2 years before diagnosis (Supplementary Figure 2).

Prediction of Onset of Crohn's Disease. At all yearly intervals before diagnosis, the titers of each antibody, based on their estimated mean abundance, were significantly higher in CD than in HC (except for p-ANCA) (Supplementary Figure 5*A*). Markers' individual performances in predicting CD can be found in Supplementary Table 3. Among the antimicrobial antibody markers, anti-flagellin X and ASCA-IgA were predictive of CD and selected by the LASSO algorithm at all timepoints before diagnosis (Figure 2*A*). The predictive performance of the LASSO model varied from 0.69 at 5 years and 0.76 at 1 year before diagnosis (Figure 2*C*).

Overall, 51 SomaLogic biomarkers were predictive of CD development (Figure 2*B*). Supplementary Table 4 shows the AUC from the 5-fold cross-validation technique for each of the selected markers. The predictive performance of these combined markers ranged from an AUC of 0.76 at 5 years to 0.88 at 1 year before diagnosis (Figure 2*D*). As shown in Supplementary Table 5, the confidence intervals of AUC were overlapping between year −5 and year −2, confirming

Table 1. Demographic and Clinical Characteristics of the Population

	Patients with Crohn's disease	Patient's with ulcerative colitis	Healthy control individuals
Number of individuals	200	199	200
Number of samples	546	599	679
Age, *y*, mean ± SD[a]	31.4 ± 6.6	28.9 ± 5.2	28.5 ± 4.8
Male sex, %	80	91	91
White race, %	69	92	91
Disease location at diagnosis, %	L1: 28 L2: 20 L3: 36 Unknown: 16 Perianal disease: 10	E1: 15 E2: 23 E3: 46 Unknown: 16	—
Disease behavior at diagnosis, %	B1: 77 B2: 18 B3: 5	—	—

NOTE. Disease location and behavior at diagnosis were categorized using the Montreal classification: L (location): L1, ileal disease; L2, colonic disease; L3, ileocolonic disease; B (behavior): B1, inflammatory phenotype; B2, obstructive phenotype; B3, penetrating phenotype; E1, proctitis; E2, left-sided colitis; E3, extensive colitis.
SD, standard deviation.
[a]Age at Sample A (HC) or diagnosis (inflammatory bowel disease).

that similar predictive performance were observed over time. However, a significant increase in predictive performance was observed at year −1 compared to previous years. The improved performance observed at year −1 was driven by 10% of selected markers whose abundance significantly increased in CD from group D and group C to group A samples (adjusted *P* value from *t* test, <15%) (Supplementary Figures 5*B* and 6). Among the selected markers, 6 SOMAmers (C-reactive protein [CRP], C5, trypsin 2 [PRSS2], serum amyloid-P [APCS], osteomodulin [OMD], and aggrecan core protein [ACAN]) were associated with an AUC greater than 0.70 across all years before diagnosis, ranging from 0.75 at 5 years to 0.81 at 1 year before diagnosis (Supplementary Figure 7*A*). Four additional SOMAmers (complement factor I [CFI], interleukin [IL] 11 receptor subunit alpha, macrophage mannose receptor 1 [MRC1], and protein SET II [SET]) involved in the immune response, cytokine-mediated signaling pathway, and apoptosis were commonly selected at timepoints 1 to 4 years before diagnosis. As shown (Figure 2*E*), the integration of the antibody markers with the SomaLogic biomarkers did not result in better predictive performance for most years before diagnosis. The median of the AUC across 100 iterations was higher than 0.75 for all years before diagnosis, confirming the high predictive performance of the selected markers (Supplementary Figure 7*C* and *D*).

Risk of Developing Crohn's Disease. For this analysis, we considered the set of 6 SOMAmers commonly selected across all timepoints before diagnosis (CRP, C5, PRSS2, APCS, OMD, and ACAN) and the best predictive antimicrobial marker: ASCA-IgA. As shown in Figure 3, the probabilities were significantly higher for patients with CD than for HC individuals across all combination of markers and timepoints considered (Bonferroni adjusted *P* values from Wilcoxon test, <.01). The inclusion of ASCA-IgA further improved the separation between the probability density of patients with CD and HC individuals (Figure 3*B*). The most predictive single markers in antibodies and SomaLogic, that is, ASCA-IgA and CRP (Supplementary Figure 8), did not perform as well as the multivariate models presented in Figure 3.

Pathway-Driven Prediction Model for Crohn's Disease. Several pathways were enriched in the set of markers predictive of CD at different timepoints before diagnosis (Figure 4*A* and Supplementary Files 1 and 2). The predictive performance of enriched pathways was validated using the training test; the top 4 predictive pathways (AUCs of >0.70) were Kyoto Encyclopedia of Genes and Genomes (KEGG) Complement Cascades, Reactome Glycosaminoglycan Metabolism, KEGG Innate Immune System, and KEGG Lysosome (Figure 4*B*).

Prediction of Onset of Ulcerative Colitis. The combination of antibody markers for UC provided lower predictive performance, with AUCs ranging from 0.57 at 5 years to 0.61 at 1 year before diagnosis (Supplementary Figure 9*A*). Their predictive performances are shown in Supplementary Table 6. The predictive performance of pANCA ranged from 0.59 to 0.61. The same statistical approach was used for UC to select protein markers associated with disease; however, their overall predictive performance was suboptimal, with an AUCs ranging from 0.49 at 5 years before diagnosis to 0.68 at 1 year before diagnosis (Supplementary Figure 9*B* and Supplementary Tables 7 and 8). SOMAmers commonly identified through 4 years before diagnosis included IL11 receptor subunit alpha and lymphocyte activation gene 3 protein (involved in antigen presentation through MHC class II). The combination of the 2 panels of markers (antibodies and SomaLogic biomarkers) slightly increased the predictive performance, although it still remained low throughout every timepoint before diagnosis (Supplementary Figure 9*C*).

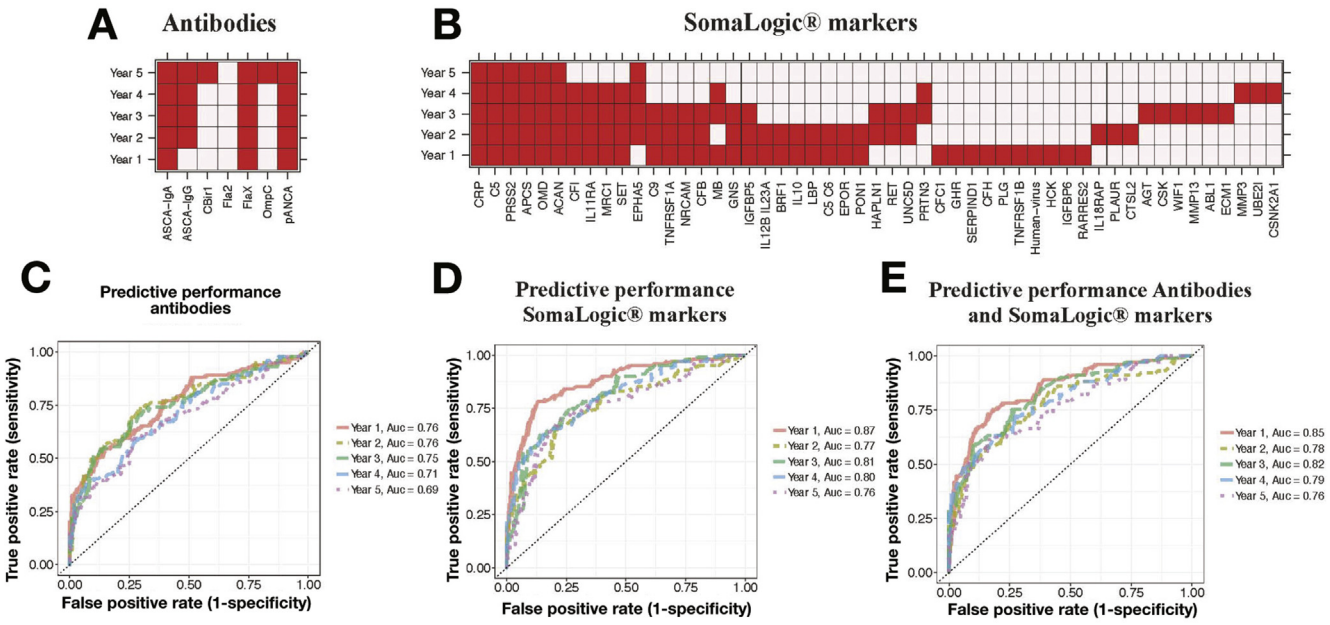


Figure 2. Prediction of CD based on different data types. (*A*) Selected antibody markers by the LASSO algorithm represented at different timepoints before diagnosis, represented by the red squares. (*B*) Selected SOMAmers based on the 5-fold cross-validation technique on the training data, represented by the red squares. (*C*) Predictive performance of selected antibody markers in terms of ROC curves for years 1, 2, 3, 4, and 5 before diagnosis. (*D*) Predictive performance of selected SOMAmers in terms of ROC curves for years 1, 2, 3, 4, and 5 before diagnosis. (*E*) Predictive performance of the integration of both antibodies and SomaLogic markers in terms of ROC curves for years 1, 2, 3, 4, and 5 before diagnosis.

Discussion

The DMSS, when integrated with the DoD-SR, is a large, unique resource that provided us the opportunity to examine the serologic progression of IBD before its clinical onset. The main observation is that it is possible to predict CD up to 5 years before diagnosis by using a panel of antibodies and protein biomarkers. Furthermore, pathway analysis based on proteomics showed that several biomarkers mapped to pathways were associated with the complement cascade, innate immunity, glycosaminoglycan metabolism, and lysosome, suggesting that these biological processes are activated many years before diagnosis.

The concept of a preclinical phase of disease in IBD was initially suggested by data from a small military cohort, where patients displayed positive ASCA or ANCA antibodies in a significantly higher proportion than control individuals years before diagnosis.[19] This was followed by 2 small studies where it was shown that the combination of multiple markers (ASCA-IgA and IgG, anti-Cbir1, anti-OmpC, and pANCA) had better predictive accuracy than each marker alone[20] and that the abundance of inflammatory biomarkers (high-sensitivity CRP and IL6) was significantly higher in patients before IBD diagnosis than in control individuals.[21] Moreover, in our previous study, using a different set of samples from the PREDICTS cohort and with no control group, we also showed that those patients with CD who develop complications at diagnosis present higher titer values and numbers of antimicrobial markers in their serum years before diagnosis.[5] These studies provided preliminary evidence, that, similar to other immune-mediated inflammatory diseases, IBD also may have a preclinical phase that might be reliably detected by disease-specific biomarkers. However, these prior studies were hampered by a small number of patients and samples, lack of longitudinal prediagnostic samples, and limited number of biomarkers investigated. Additionally, predictive accuracy was not evaluated on any validation set, and therefore, these studies were not able to assess whether the selected model incurred overfitting. Here, we have successfully overcome these prior limitations. Leveraging our access to multiple prediagnosis samples up to 5 years before clinical diagnosis in a well-characterized cohort followed over time and our ability to test a large array of markers and handle high-dimensional data, we were able to rigorously study and internally validate the temporal association of biomarkers with the disease of interest. In addition, the functional role in disease pathogenesis of the selected proteomic markers was assessed through comprehensive pathway analysis.

Using the SomaLogic platform, we have, for the first time to our knowledge, identified a panel of highly sensitive candidate protein biomarkers for the prediction of CD, with a6% accuracy at 5 years before diagnosis and 88% accuracy at 1 year before diagnosis. Overall, 51 protein biomarkers were found predictive of CD, with 57% overlapping at least 2 different timepoints. The most consistent proteomic markers preceding disease diagnosis were markers of inflammation (CRP, serum amyloid P, trypsin 2), markers involved in cytokine signaling, innate immunity, and response to bacteria (complement factors, tumor necrosis factor receptor subunits, lipopolysaccharide binding protein, proteinase-3, and several interleukins). Antimicrobial markers were also significantly higher in preclinical samples

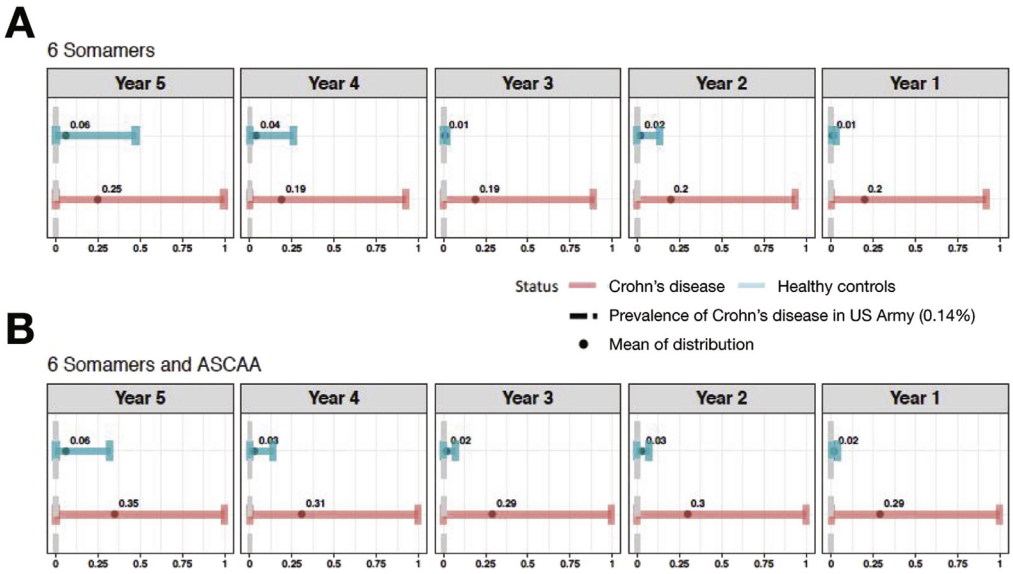


Figure 3. Confidence intervals of the probability of developing Crohn's disease at years 1, 2, 3, 4, and 5 before diagnosis for HC individuals and patients with CD given different sets of markers. (*A*) The 95% confidence intervals of the probability of developing the disease given the 6 SOMAmers selected at every year before diagnosis and (*B*) the 95% confidence intervals of developing Crohn's disease given the 6 SOMAmers and the antimicrobial marker ASCA-IgA. For each confidence interval, the mean is shown by a black dot, and the prevalence of CD in the US Army population (0.14%) is shown as a gray vertical line.

from patients with CD compared to control individuals. The titers of these markers were quite stable before diagnosis, as shown by their time-varying trajectories. Although the timing and instigating factors for the generation of these biomarkers remain largely unknown, it is thought that they might represent an aberrant adaptive immune response to environmental triggers and/or commensal microbiota.[22,23] They are associated with disease phenotype and risk of developing complications, but not with disease activity, which may explain their stability over time.[24] Finally, the combination of selected markers across several years before diagnosis yielded an excellent discriminative probability of developing CD as compared to HC individuals, especially when combined with ASCA-IgA.

The ultimate goal of disease prediction is to be able to identify individuals at high risk for disease development in whom preventive and disease-attenuating strategies could be implemented. It is plausible that IBD is preceded by a

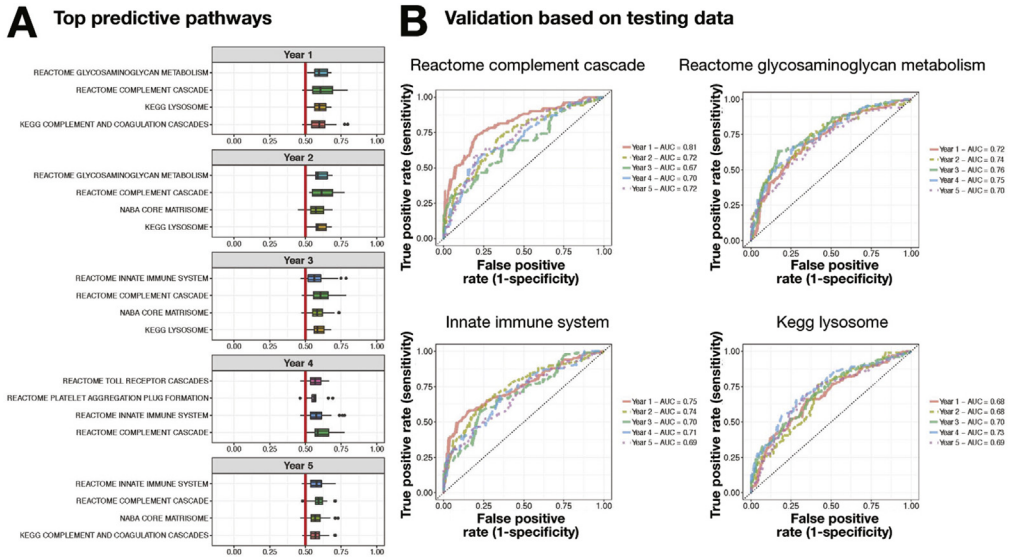


Figure 4. (*A*) Top predictive pathways for CD at years 1, 2, 3, 4, and 5 before diagnosis based on training data. For each year before diagnosis, the top 4 pathways with the smallest *P* values are displayed. For each pathway, the boxplot of AUC values for the group of SOMAmers mapping to the pathway is shown. AUC values are based on the 5-fold cross-validation technique based on the training data alone. (*B*) SOMAmers mapping to each pathway were used in a multivariate model to predict CD for samples in the testing data. ROC curves for different years before diagnosis are shown.

preclinical phase, where, in the absence of symptoms and bowel lesions, deregulated immune pathways, dysbiosis, and other key pathogenic mechanisms are already in place, leading to the expansion of the autoinflammatory process that will eventually culminate in subclinical intestinal inflammation, tissue injury, and, finally, symptoms, culminating in a diagnosis. This preclinical phase is of unknown duration but conceivably could start many years before diagnosis. It is know that early intervention in CD is associated with better outcomes[25]; therefore, it is conceivable that interventions introduced during an even earlier preclinical phase of disease could more easily abrogate or modulate the dysregulated immune responses. However, it remains unclear which interventions could offer the potential to ameliorate or reverse the immunologic changes in this phase. Once disease is established, it becomes difficult to distinguish between the pathways that may be responsible for disease initiation from those resulting from the dysregulated cycle of inflammation. We hypothesized that performing pathway analysis on the proteomic biomarkers with the highest predictive performance for IBD with cross checking against multiple well-annotated databases might identify candidate pathways that have biologic plausibility in preclinical IBD pathogenesis. Indeed, by using the SOMAscan proteomic assay spanning a broad array of biological processes, we were able to identify several dysregulated pathways in preclinical CD that could be informative for preventive and therapeutic interventions. The lysosome pathway, deeply interrelated with autophagy, one of the main defective processes in IBD, was significantly enriched in patients with CD in comparison with control individuals, especially at 4 years before diagnosis.[26] Pathways involved in glycosaminoglycan metabolism were enriched in patients with pre-CD as compared to HC individuals, which is interesting given the accumulating evidence linking defects in glycosylation to inflammation and IBD. Glycans are implicated in the mechanisms underlying the immunopathogenesis of IBD, through modulation of the threshold for T-cell activation,[27] and recently, it has been shown that the glycosylation profile of IgG is associated with the clinical severity of the disease,[28] further adding to a potential role of glycans in the immunopathogenesis of IBD. The most predictive pathway of CD development was the complement pathway, known to be crucial for innate antimicrobial defense, recognizing and responding locally and systemically to microbial-associated molecular patterns and triggering appropriate immune responses of major importance for the maintenance of intestinal barrier function.[29] These findings, combined with the presence of antimicrobial antibodies years before diagnosis, collectively support the hypothesis of altered host-microbiome interactions as one of the earliest events in the pathogenesis of CD. We can hypothesize that acquiring this type of mechanistic information in preclinical disease can potentially uncover key pathogenic events and lead to the development of new therapeutic targets. We therefore speculate that based on our findings, microbiome-directed therapies or interventions, such as dietary interventions; prebiotic, probiotic, or antibiotic use; phage therapy; fecal transplantation; synthetic microbiome treatment; or other original ways of modulating and reprogramming the gut flora, could be implemented during the preclinical phase of disease, potentially as a preventative strategy.[4]

Interestingly, and in contrast to CD, no single marker alone or in combination provided good predictive performance for UC, especially in the years more distant from the time of diagnosis. Specifically, pANCA, the most specific marker for UC, had a suboptimal performance, especially in the most distant years; this fits with the clinical observation that usually UC has a more abrupt clinical presentation and may imply a shorter preclinical period; alternatively, different biological pathways underlying disease initiation and not captured in our biomarker platform may exist. Being a mucosal disease, it is also plausible that mucosal or stool biomarkers could be more appropriate for the prediction or earlier detection of disease.

Our work has several strengths and limitations. We used prediagnostic samples from a well-characterized cohort, allowing for a particularly rigorous analysis and internal validation within our cohort. Having access to multiple prediagnosis timepoints allowed us to model the time-varying trajectories by using a stochastic process able to capture the change along time of each marker. Furthermore, we measured a wide selection of markers and used a novel proteomic platform, which, for the first time to our knowledge, allowed us to identify putative biological pathways enriched in preclinical disease. Our study is not without some limitations, particularly reliance on serum only, imperfect demographic matching, and generalizability, given that it was restricted to young, predominantly male, active duty military personnel. Some covariates, including tobacco use and family history, known to modulate disease risk were not available. Furthermore, it is possible that in some patients with CD, disease was already established but not recognized because of the delayed diagnosis or absence of symptoms; however, the prediction of markers 5 years before diagnosis likely extends well beyond the reported diagnostic delay and the undetected clinical phase of disease.[30] Likewise, it is also conceivable that a small proportion of HC individuals, just based on the military population disease incidence, may develop IBD in the future. Finally, the biomarkers identified and the pathway analysis may have been limited by the background chosen from SomaLogic, enriched in markers involved in inflammation and immunity. Because we excluded other immune-mediated diseases from the HC population, we cannot assess the predictive accuracy of the set of biomarkers identified in these conditions; however, the fact that autophagy, a CD-specific process, was found to up-regulated suggests that our biomarkers maybe disease specific. The addition of antimicrobial markers (not usually present in other immune-mediated conditions) to the predictive tool, even if its overall predictive accuracy was lower than SomaLogic alone, further improves the marker's specificity.

In summary, our data suggest CD may be predicted years before diagnosis, but we could not identify predictors for UC. We believe our work represents an important foundation for future work in the preclinical phase of disease. Our

results should be validated in independent populations, and the performance of our predictive tool should be improved with genetic, microbiome, and relevant clinical data (eg, smoking, family history) obtained from ongoing longitudinal cohorts. Such multidimensional predictive tools could be used in prospective studies or disease prevention trials to select a priori populations at high risk for developing IBD. IBD prediction and prevention may be a possibility in the future, and it is clear that further research in this field is warranted. In the future, early identification of individuals at high risk for disease development could allow for close monitoring and for the development of interventions to delay, attenuate, or even halt disease initiation.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at https://doi.org/10.1053/j.gastro.2020.03.007.

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Conflicts of interest
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